Datathon 2026: Part 2 Preview

1 Exploratory Data Analysis & Strategic Prescriptions

1.1 Data Story 1: Channel Conversion Efficiency Is Eroding While Return Risk Remains Structurally Flat

Descriptive. Joining monthly *web_traffic* and *orders* shows wide efficiency dispersion: social media leads at 10.40 orders per 1,000 sessions, while organic search records 7.39. Post-purchase quality is stable, with return rates tightly bounded between 3.33% and 3.46%.

Diagnostic. The gap is a top-of-funnel productivity problem, not a fulfillment-quality problem. High-traffic but low-yield channels (notably organic and email) depress blended media productivity despite high absolute revenue.

Predictive. Monthly trend slopes for *orders_per_1k_sessions* are negative in every channel (-0.123 to -0.093), implying rising CAC pressure if the acquisition mix remains unchanged.

Prescriptive. Reallocating 100,000 sessions from the least efficient to the most efficient channel yields +300.84 incremental orders, about +7.29 million net sales, and +0.71 million gross profit under current unit economics. Implement a weekly allocation rule: scale spend only when trailing 4-week *orders_per_1k_sessions* is above channel median and *return_rate* is below portfolio p75.

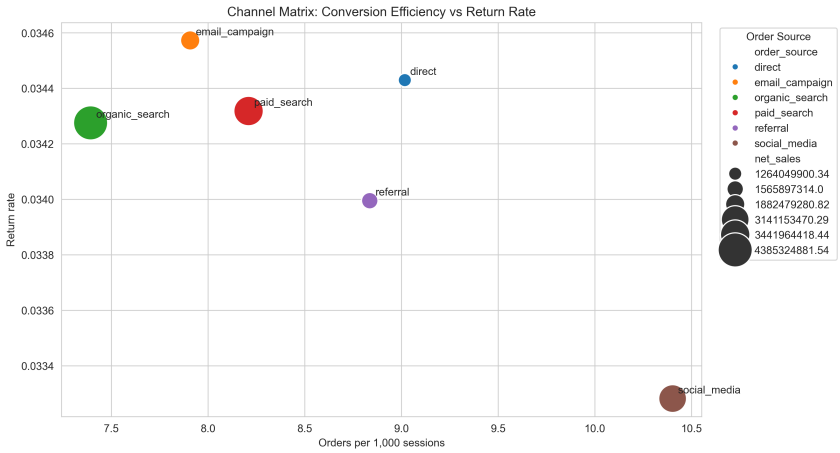

Figure 1: **Channel Efficiency–Quality Frontier.** Bubble size denotes channel net sales; the x-axis reports conversion efficiency (orders per 1,000 sessions) and the y-axis reports return rate. Conversion heterogeneity is large while return risk is nearly flat, supporting short-run growth via traffic reallocation toward high-yield channels rather than quality remediation.

1.2 Data Story 2: Discount Intensity Beyond 10% Systematically Destroys Gross Margin

Descriptive. Margin by realized discount bucket is strongly non-linear: $\leq 5\%$ remains positive (+17.24%), 5–10% is near break-even (+0.87%), while 10–20% (-10.85%) and $> 20\%$ (-24.56%)

are negative. Promotion-type results are consistent: no-promotion +19.96%, percentage -10.25%, fixed-amount -63.28%.

Diagnostic. The main leak is not promotion incidence itself, but deep-discount design and fixed-value mechanics that decouple markdown depth from item economics. This generates negative-contribution demand, especially when high-discount share rises in peak campaigns.

Predictive. If the deep-discount mix persists, high-volume campaign periods will keep compressing blended gross margin and increase earnings volatility, even with stable topline.

Prescriptive. Set a default markdown cap at 10%. Permit $> 10\%$ discounts only for SKUs in the top quartile of *days_of_supply* and with non-negative projected post-discount contribution. A counterfactual re-mix shifting 20% of net sales from $> 10\%$ buckets to 5–10% yields about +96.06 million incremental gross profit.

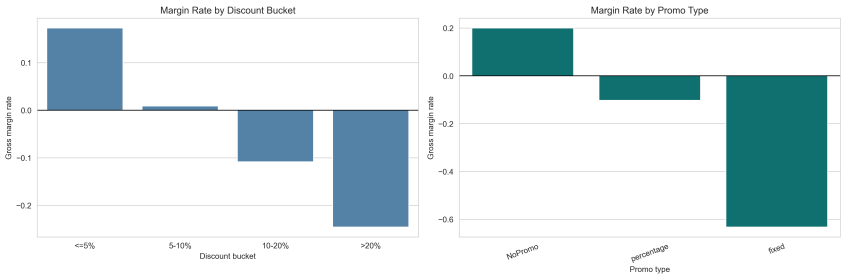

Figure 2: **Profitability Response to Discount Architecture.** The left panel reports gross margin by discount bucket; the right panel reports gross margin by promotion type. The margin sign-flip beyond 10% and the severe underperformance of fixed-value promotions directly support discount guardrails and contribution-based campaign approval.

1.3 Data Story 3: Joint Overstock and Refund Exposure Is Locking Working Capital

Descriptive. Integrating latest inventory snapshots with category-level returns shows concentrated capital lock-in. Streetwear has the largest inventory cost (about 276.37 million), with 82.65% overstock and a 34.89% refund-to-gross-profit ratio. Casual has the highest refund burden relative to gross profit (41.57%), while GenZ has the highest overstock rate (84.09%).

Diagnostic. The risk reflects a two-sided operating failure: excess stock depth in slow-turn nodes and weak post-purchase retention in selected categories. Categories in the matrix upper-right simultaneously consume working capital and erode realized profitability.

Predictive. Without intervention, persistent overstock and refund leakage will increase cash-conversion-cycle pressure, reduce inventory agility, and crowd out reinvestment in high-velocity assortments.

Prescriptive. Prioritize Streetwear and GenZ for integrated liquidation-plus-quality programs with explicit margin floors. For Streetwear alone, a 10–15% refund reduction protects 40.67–61.01 million in value; reducing locked inventory value by 5 percentage points releases about 13.82 million working capital.

1.4 Executive Synthesis

Across all three stories, the optimization axis is not revenue at any cost, but contribution-quality growth under capital discipline along *Channel* $\times$ *Discount Depth* $\times$ *Inventory Risk*. We recommend a weekly control-tower tracking five KPIs: *orders_per_1k_sessions*, *gross_margin_rate*, *return_rate*, *overstock_rate*, and *refund_to_gp*, with explicit trigger thresholds linked to budget, markdown, and replenishment decisions.

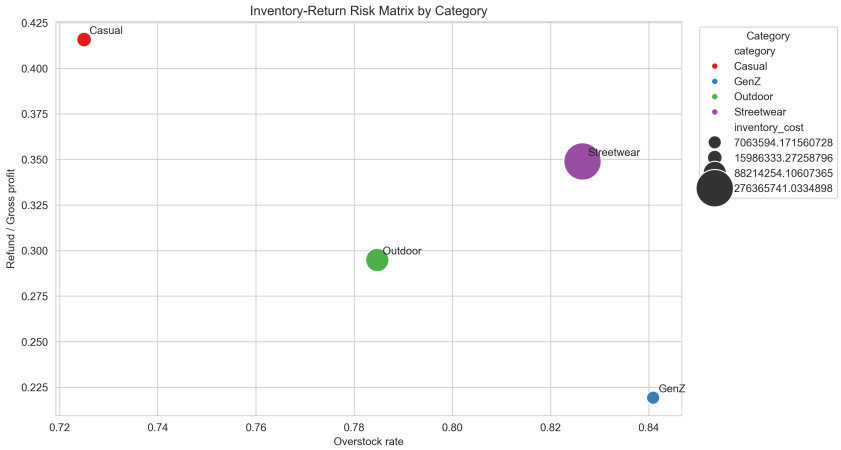

Figure 3: **Inventory–Refund Risk Matrix by Category.** Bubble size encodes latest inventory cost. Categories in the upper-right quadrant combine high overstock rates with high refund-to-gross-profit ratios, signaling concurrent liquidity lock-in and profit leakage. The plot isolates where coordinated pricing, inventory normalization, and return mitigation produce the largest near-term financial impact.